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Cation-Exchange Doping of n-Type Conjugated Polymers Entailing Glycolated Side Chains

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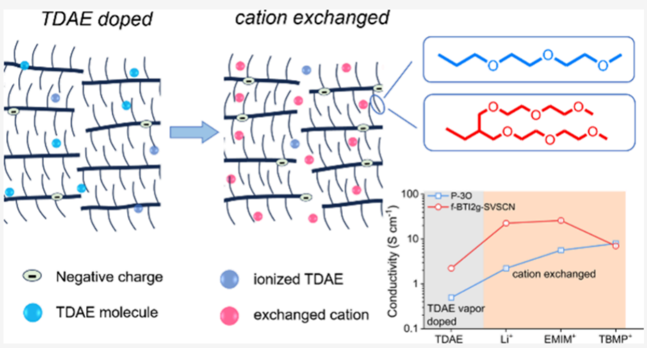
ABSTRACT: Improving the doping efficiency of n-type conjugated polymers is a significant challenge in the field. However, the limited availability of dopants and methods for doping result in the electrical performance of n-type materials lagging behind that of p-type materials. In this study, we introduce a novel method for n-doping called cation-exchange doping, which enhances the doping levels of n-type polymers that incorporate glycolated side chains. We selected two n-type conjugated polymers to undergo cation-exchange doping using the direct dopant tetrakis-(dimethylamino)ethylene (TDAE), along with three cations Li^+, 1-ethyl-3-methylimidazolium, and tributylmethylphosphonium. All of these doped materials demonstrate improved thermoelectric performance and achieve deeper doping levels compared with doping with TDAE alone. The two polymers exhibit distinct electrical properties and thermoelectric performance related to cations, highlighting the importance of counterion engineering in optimizing the outcomes of cation-exchange doping. This can be explained by the cation-mediated density of states and electrical conductivity resulting from electrochemical doping. This research demonstrates a universal method for cation-exchange doping in n-type polymer semiconductors and offers new insights into understanding the interaction between polymers and counterions in organic electronics.



1. INTRODUCTION

Organic semiconductors (OSCs) are extensively researched for applications such as light-emitting diodes,[1−3] solar cells,[4−6] and transistors.[7−9] To enhance the optoelectrical properties of OSCs, doping with certain molecules is essential as it can significantly alter the Fermi levels of host semiconductors.[10−12] Doping efficiency is a vital factor in this process: an excess of dopants in the host polymer can lead to issues like severe phase segregation,[13−15] electron scattering,[16,17] and the formation of Coulombic traps.[18,19] Chemical doping involves a redox reaction between the polymer and the dopants, driven by the energy offset between the donor and acceptor materials.[20,21] Two models are proposed to explain this doping process: a) integer charge transfer (ICT) and b) hybridization of frontier orbitals.[21−23] ICT is preferable due to its higher doping efficiency. Therefore, carefully designing π-conjugated materials and selecting appropriate dopants are crucial for achieving proper energy level alignment and improving the doping efficiency.[24−26] While there has been notable progress in enhancing electrical performance through p-doping, there are still limited doping methods and stable air-compatible dopants available, which restrict the development of n-doping properties.[27−29]

One persistent challenge in advancing n-doping is the poor miscibility between dopants and the host material, which limits the doping efficiency.[30,31] A universal strategy to address this issue is the introduction of ethylene glycol-type side chains to π-conjugated polymers.[32−34] The polar ethylene glycol side chains enhance the surface morphology of the dopant/polymer mixture, thereby improving the doping efficiency. Additionally, the careful design of ethylene glycol side chains can significantly shift the position of the reacted dopant molecule away from the conjugated backbone, which in turn modulates the Coulombic interactions between the counterion and the carrier.[35,36] Despite the advantages of ethylene glycol side chains, a more general approach to further enhance the electrical performance of n-conjugated polymers is still needed.

The ion-exchange method presented by Watanabe's group has been proven to be an effective approach for enhancing electrical performance and has been extensively studied in the

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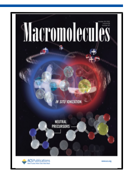

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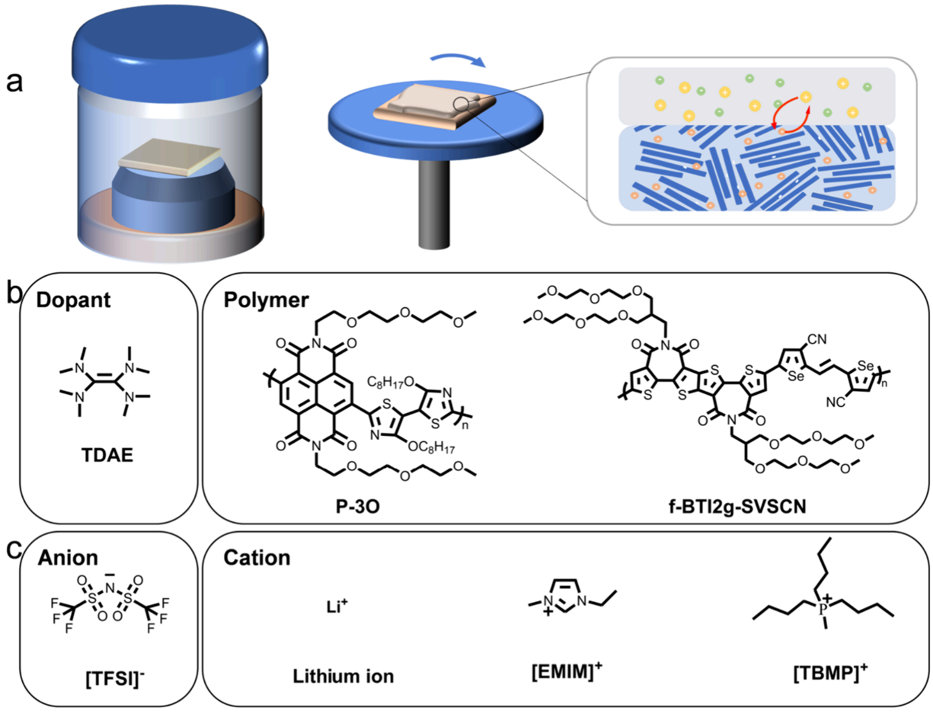

Figure 1. (a) Cation-exchange doping process. The chemical structures of (b) dopant TDAE, polymer P-3O, and f-BTI2g-SVSCN, and (c) selected anion and cations used for cation exchange (dissolved in ACN).

realm of p-type doping.[37] Remarkably, a state-of-the-art electrical conductivity of over 1000 S cm^{-1} has been achieved in cation-exchange doped poly[2,5-bis(3- tetradecylthiophen-2-yl)thieno[3,2-*b*]thiophene].[38,39] The notable electrical performance observed in p-doping through anion exchange highlights the substantial potential to improve the usually lower performance of n-doping achieved through cation exchange. In general, there exists a chemical equilibrium between the neutral state and the doped state in p-doped films:

$$\text{(neural state)[polymer]} + \text{dopant} \rightleftharpoons [\text{polymer}]^{+} + \text{dopant}^{-}\text{(doped state)}$$

The ion-exchange process involves two steps: (1) doping an organic semiconductor with a direct dopant such as $FeCl_3$ for p-type doping and (2) replacing the dopant counterion present in the polymer films with other ions at the solid/liquid interface.[40] These two steps occur consecutively, and the substitution of the ionized dopant shifts the chemical equilibrium of the redox reaction between the dopants and the host polymers toward a deeper doped state. This approach significantly enhances the redox strengths of the dopants beyond the limitations of the energy levels.[41] Furthermore, the careful modulation of the counterion in the polymer films influences the aggregate structure and the Coulombic interactions with transport carriers, which reduce the energetic disorder and the charge-transport barrier for doped semiconductors.[42] Cation exchange has also recently been employed in n-doping systems such as P(NDI2OD-T2) and P(PzDPP-2FT), which entail alkyl side chains.[43,44] This process realizes simultaneous doping and cation exchange, further enhancing the electrical conductivity and ambient stability through ongoing cation exchange. However, the mechanism of this powerful doping strategy remains elusive, and its validity for conjugated polymers with glycolated side chains, which enhance the doping efficiency, requires further verification.

Herein, we introduce the novel n-doping method, cation exchange, to further promote the electrical performance and doping efficiency of the n-type conjugated polymer entailing glycolated side chains. Two conjugated polymers incorporating glycolated side chains with different molecule packing orientations, namely, P-3O and f-BTI2g-SVSCN, are investigated with a commercial n-dopant tetrakis(dimethylamino)-ethylene (TDAE) by vapor doping. We conducted ultraviolet–visible-infrared (UV–vis-IR) spectroscopy, Fourier transform infrared spectroscopy (FT-IR), etc. to verify the occurrence of cation exchange and determine the doping level. Both polymers' cation-exchange doping [Li^+, 1-ethyl-3-methylimidazolium ($EMIM^+$), and tributylmethylphosphonium ($TBMP^+$)] exhibited enhanced electrical conductivities and thermoelectric performance compared to TDAE vapor doping. The measurement of the density of state of the polymer and the electrochemically modulated electrical conductivity explains the origin of cation-related electronic properties of conjugated polymers.

2. EXPERIMENTAL SECTION

2.1. Materials. The conjugated polymers P-3O and f-BTI2g-SVSCN (chemical structures in Figure 1) were synthesized according to the reported procedures.[33,45] The n-dopants TDAE and three kinds of ionic salts (LiTFSI, EMIMTFSI, and TBMPTFSI) [$TFSI^-$: bis(trifluoromethylsulfonyl)imide)] were purchased from Sigma-Aldrich.

2.2. Device Fabrication. The glass substrates were cleaned with a commercial detergent, deionized water, acetone, and isopropanol. Then, the glass substrates were dried in a 100 °C oven and ultraviolet ozone-treated before use. P-3O and f-BTI2g-SVSCN were dissolved in 5 mg/mL chloroform and hexafluoroisopropanol, respectively, and stirred overnight. The polymer films were deposited with a spin-coating speed of 2000 rpm for 40 s. The films were then annealed at

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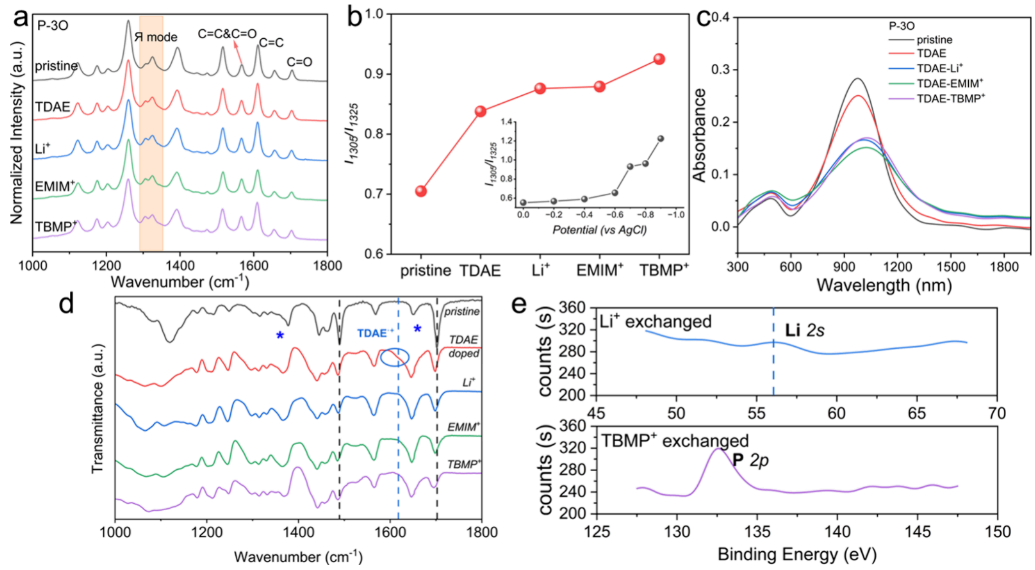

Figure 2. (a) Raman vibrational spectra (excited at wavelength of 532 nm), (b) variation of Raman peak ratio I_{1305}/I_{1325} (the inset figure is the relative intensity change of I_{1305}/I_{1325} with the increasing doping potential versus AgCl), (c) UV−vis-IR spectra, and (d) FT-IR transmittance spectra of pristine, TDAE doping, and cation exchanged P-3O. The star label denotes doping-induced features. Gray and blue broken lines correspond to FT-IR transitions of the pristine polymer and $TDAE^{+}$, respectively. (e) XPS of Li^{+} and $TBMP^{+}$ exchanged P-3O films.

110 °C for 30 min and 150 °C for 30 min in an inert atmosphere. The film thickness was near 30 nm which were defined by a KLA Tencor P7 instrument.

2.3. TDAE Vapor Doping and Cation Exchange. For the TDAE vapor doping, 50 μL of TDAE was added into a 50 mL glass jar. The polymer film was placed into the jar which was heated at 50 °C and doped for different times. The cation exchange of polymer films was executed as follows: First, the polymer films were vapor-doped by TDAE for 5 min. The doped polymer films were cation exchanged with 200 mM ion salt solution in acetonitrile (ACN) for 30s (LiTFSI, EMIMTFSI, and TBMPTFSI). Then, the extra solution was spun out of the film with 2000 rpm for 30s. This process was repeated three times, and 300 μL of ACN was used to wash the residual ion salts with a speed of 2000 rpm. The control films of TDAE doping 5 min were also washed by 300 μL of ACN, although the electrical conductivity was nearly unchanged.

2.4. Detection of Cation Exchange. The Raman spectra were recorded with a LabRam HR 800 laser confocal Raman spectrometer. The fabricated devices on the Au/glass substrate were excited at 532 nm (integral time 10 s) while recording the Raman signal. For the electrochemically doped Raman spectra, Ag/AgCl was employed as a pseudo-reference electrode where 0.1 M NaCl aqueous solution was used as the electrolyte. For each doping potential, a period time of 120 s was employed to ensure the polymer films were entirely doped. For the UV−vis-IR spectra collected by Shimadu UV-3700 spectrophotometers (Japan), the P-3O and f-BTI2g-SVSCN films were prepared and sealed by a glass. FT-IR spectroscopy was performed with a Bruker Alpha FT-IR spectrometer in the transmission mode. The polymer solution was drop-cast onto the KBr tablet in an inert atmosphere. X-ray photoelectron spectroscopy (XPS) measurements were performed with an ESCALAB-MKII 250 photoelectron spectrometer (VG Co.). XPS was performed using Mg Kα (1200 eV) as the excitation source.

2.5. Electrical and Thermoelectric Performance Characterization. Electrical conductivity measurement: The electrical conductivity (σ) was measured in a N_2 filled glovebox by a Keithley 2450 instrument controlled by self-coded Labview. The four-probe electrical conductivity was calculated as $\sigma = (I/V) \times l/(w \times d)$, for which the channel length (l) is 0.5 mm, the channel width (w) is 3.0 mm, and the film thickness (d) is near 40 nm. For each doping concentration, a set of six devices were recorded. Seebeck coefficient measurement: The Seebeck coefficient (S) was also measured in an inert atmosphere. A set of Peltier was employed to serve as the temperature gradient. During measurement, the temperature variation and voltage difference were, respectively, monitored by DAQ970A and DAQ973A. For each doping concentration, a set of three devices were recorded.

2.6. Other Characterizations. The EPR spectra were monitored with a Bruker EMXnano EPR spectrometer. The X-band EPR spectra of polymers were recorded at 9.615 GHz with a modulation amplitude of 0.02 mT and a microwave power of 1 mW. AFM was performed with a Bruker Dimension ICON atomic force microscope in the peak force mode. The scan rate was 1.0 Hz, and each sample contained 256 lines. The peak force set point was 0.02 V. Grazing incidence wide angle X-ray scattering (GIWAXS) data were obtained at 1W1A Diffuse X-ray Scattering Station, Beijing Synchrotron Radiation Facility (BSRF-1W1A). Density of states (DOS) measurements were obtained by using a three-electrode electrochemical configuration. The working electrode comprised spin-cast polymer films deposited on Au substrates, while the reference and counter electrodes consisted of Ag/AgCl and Pt wire, respectively. Linear sweep voltammetry was performed at a scan rate of 1 mV s^{-1} under inert conditions. Gate voltage-modulated electrical conductivity measurements employed an interdigitated electrode array with channel width (w = 400 μm) and length (l = 20 μm) serving as source/drain contacts, while a AgCl electrode applied gate bias (device schematic shown in Figure 5). Conductivity was calculated using $\sigma = I_{DS}/V_{DS} \times l/wd$, where V_{DS} = 10 mV and d represents the film thickness.

3. RESULTS AND DISCUSSION

3.1. Cation Exchange and Chemical Structure. The two-step cation-exchange doping of an n-type semiconductor is illustrated in Figure 1a. In this process, cation-exchange occurs after vapor doping with TDAE. Separating these processes allows for a better understanding of the interactions between carriers and counterions, independent of the carrier density. Figure 1b shows the chemical structures of dopant TDAE, n-type polymer P-3O, and f-BTI2g-SVSCN, which feature glycolated side chains. The structure of P-3O is a straight chain, whereas f-BTI2g-SVSCN has a branched glycolated chain. It is worth mentioning that these two polymers adopt different molecule packing orientations (edge-on for P-3O and

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face-on for f-BTI2g-SVSCN, as shown in Figure S1). The importance of this different aggregating structure will be discussed later. Both materials have previously been investigated in organic thin-film transistors and organic electrochemical transistors.[33,45] The selected cations Li^+, $EMIM^+$, and $TBMP^+$ used for exchange are also displayed in Figure 1c. They differed in polarity and ion size. The anion $TFSI^-$ which is stable for the TDAE doping served as a common anion for using the three cations during the cation-exchange doping. Herein, a TDAE vapor doping time of 5 min was adopted for cation exchange.

3.2. Detection of Cation Exchange. Vibrational Raman spectroscopy of P-3O, illustrated in Figure 2a, was conducted to investigate the changes in the vibrational features induced by ion exchange. The Raman vibrational spectra of NDI-thiazole-based conjugated polymers remain scarcely studied. Given the structural similarity of the polymer backbone to P(NDI2OD-T2), the Raman vibrational band assignments were made with reference to previous studies on P(NDI2OD-T2). The peaks at 1708 and 1612 cm^{-1} are attributed to the C═O and C═C stretching vibrations, respectively, which are primarily localized on the NDI unit.[46] The peak localized at 1568 cm^{-1} corresponds to a "coupling" mode of C═C and C═O stretching (denoted as C═C and C═O) on NDI. The triplet peaks from 1220 to 1420 cm^{-1} are denoted as indicative bands of Я mode (collective C═C/C−C bonds stretching/shrinking) among the whole backbone,[47] which is quite sensitive to molecular doping. To get a better knowledge of the Raman spectra with doping level, we monitored the vibrational spectra of P-3O, which was electrochemically doped in NaCl aqueous solution (Figure S2) and demonstrated the relation of the peak ratio of I_{1305}/I_{1325} (Я-mode on NDI-2Tz) with the doping degree. As displayed in the inserted graph, the ratio of I_{1305}/I_{1325} was continuously enhanced with increased doping potential (the normalized spectra can be seen in the Supporting Information). Herein, with TDAE vapor doping, the C═O stretching attenuated, and the coupling mode of "C═C" and "C═O" intensified, indicating effective doping of P-3O. Moreover, the peak ratio I_{1305}/I_{1325} of the Я mode is also intensified with TDAE doping, which might correspond to the variated vibrational mode of the polymer backbone. After the cation exchange with Li^+, $EMIM^+$, and $TBMP^+$, no new vibrational mode emerged. The C═O stretching and coupling mode of C═C and C═O were further attenuated and strengthened, respectively. As shown in Figure 2b, the indicative peak ratio of I_{1305}/I_{1325} continuously increased with the cation varying from Li^+ to $EMIM^+$ and $TBMP^+$ (corresponding to a doping potential of −0.6 to −0.7 V), indicating the gradual doping level of P-3O by considering the results from electrochemical doping.

UV−vis-IR spectroscopy was employed to verify the validity of the cation exchange of the three ions, as shown in Figure 2c. With TDAE vapor doping for 5 min, the neural peak at 970 nm bleached. The broadened peak might result from the overlapped polaron absorption band at ∼1200 nm with the neural one.[25,48] The neural peak was bleached continuously after cation exchange with three different cations. Interestingly, the $TBMP^+$-exchanged film displayed the most intensified neural band among the three cations, which seemed contradictory to the Raman spectra's conclusion. We proposed that the peculiar trends in the optical spectra may be attributed to the different extinction coefficients induced by the various kinds of counterions.[49] We designed an experiment to verify this: an additional ionic liquid with a molar ratio of 1:4 (ionic liquid: polymer unit) was added into the 5 wt % N-DMBI doped P-3O film, and the absorption spectra were recorded. As shown in Figure S5, it was found that the type of added cation (Li^+, $EMIM^+$, and $TBMP^+$) influenced the absorption spectra of 5 wt % N-DMBI-doped P-3O accordingly. All of them increased the neutral absorption band compared to the 5 wt % N-DMBI doped P-3O film. The TBMPTFSI/N-DMBI mixed doping film showed the most intensified neutral peak absorption, and the LiTFSI/N-DMBI mixed doping film displayed the lowest intensity. This explains why the cation-exchanged films exhibit peculiar optical absorption intensity as the surrounding cations might influence the vibronic transition. We have also employed FT-IR to evaluate the exchange behavior during cation exchange, as shown in Figure 2d. The vibrational stretches at 1708 cm^{-1} (C═O) and 1492 cm^{-1} exhibited bleaching, and the doping-induced features intensified with TDAE doping and further cation exchange, indicating progressive doping. The reaction product $TDAE^{\cdot+}$ displayed a characteristic absorption at ∼1622 cm^{-1} (Figure S4), which appeared as a shoulder absorption band in the TDAE-vapor-doped P-3O. This peak nearly disappeared in the FT-IR spectra of cation-exchanged P-3O films, indicating an exchanging efficiency of nearly 100% for all of the cations.

XPS is a sensitive method that directly monitors film's characteristic elements. Here, in the three kinds of cations, lithium and phosphorus might be directly detected if there is cation exchange at the solid/liquid interface. The results are displayed in Figure 2f; there is no significant signal of Li, which might be attributed to the low sensitivity of lithium.[50] However, the $TBMP^+$-exchanged P-3O displayed a characteristic 2p signal of phosphorus element at 132.5 eV. The integral of phosphorus abundance corresponds to a density of 2×10^{20} cm^{-3} for $TBMP^+$, which corresponds to a doping level of 25%.

Similarly, Raman, UV−vis-IR, and FT-IR spectroscopies were also employed in the investigation of f-BTI2g-SVSCN, which can be found in the Supporting Information (Figures S3 and S4). By combining the results from optical and Raman spectra, it was deduced that the doped f-BTI2g-SVSCN films exchanging with $EMIM^+$ exhibit the deepest doping degree and the counterpart of $TBMP^+$ exhibits the lowest doping degree. According to the results from FT-IR, the signal of $TDAE^{\cdot+}$ still existed in the cation exchange f-BTI2g-SVSCN where the $TBMP^+$ exchanged films displayed the most $TDAE^{\cdot+}$. It demonstrates a low exchange efficiency in f-BTI2g-SVSCN, especially for $TBMP^+$. This might be correlated with a more ordered crystalline structure and polar side chain with a higher ethoxy chain of f-BTI2g-SVSCN content, which prevents the penetration of $TBMP^+$ into the polymer films.

3.3. Thermoelectric Performance of Cation-Exchange Doping. The TDAE vapor-doped P-3O and f-BTI2g-SVSCN show considerable thermoelectric performance, as shown in Figure S6. The electrical conductivity increased up to a plateau for both P-3O (5.5 S cm^{-1}) and f-BTI2g-SVSCN (36.6 S cm^{-1}) with doping times exceeding 20 and 2 h, respectively. The varying doping times required to achieve plateau conductivity may be attributed to the differing reactivity with TDAE. Additionally, the Seebeck coefficient showed a gradual decline as the doping time was extended. Ultimately, a peak power factor of 14.5 $\mu W\ m^{-1}\ K^{-2}$ was achieved for P-3O, while f-BTI2g-SVSCN reached a peak power factor of 32.3 $\mu W\ m^{-1}\ K^{-2}$.

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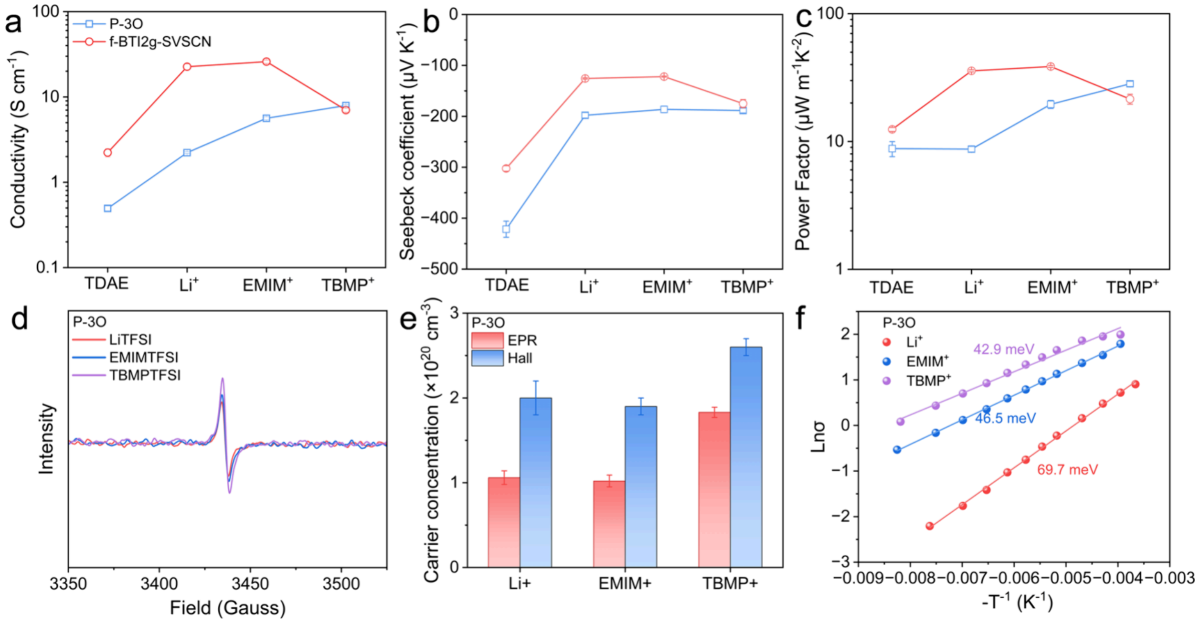

Figure 3. (a) Four-probe electrical conductivity, (b) Seebeck coefficient, and (c) calculated power factor of TDAE and cation exchanged P-3O and f-BTI2g-SVSCN films. (d) EPR spectra, (e) calculated carrier concentration, which were detected by EPR and FastHall, and (f) temperature-dependent electrical conductivity of Li^+, $EMIM^+$, and $TBMP^+$ exchanged P-3O films.

We set the TDAE vapor doping time as 5 min for all routines with and without cation exchange. A concentration of 200 mM LiTFSI, EMIMTFSI, and TBMPTFSI in ACN was employed. As shown in Figure 3a, all three exchanged films showed enhanced electrical conductivity for P-3O (2.2, 5.6, and 7.9 S cm^{-1} for Li^+, $EMIM^+$, and $TBMP^+$, respectively) compared to the counterpart doped by TDAE alone (0.5 S cm^{-1}), which indicates the successful implementation of cation-exchange doping. The Seebeck coefficient, which is determined by the energy difference between the transport energy level and the Fermi energy level,[51] reduced from −420 to ∼ −190 μV K^{-1} with cation exchange (Figure 3b). The decreased Seebeck coefficient indicated an enhanced doping degree by cation exchange,[52] consistent with previous Raman and optical spectra data. Notably, the counterion $TBMP^+$ exchanged P-3O displayed the best thermoelectric performance (PF = 28.3 μW m^{-1} K^{-2}) among the three cations, nearly 4 times larger than the TDAE doping. The substitution of $TDAE^{\cdot+}$ by cations (Li^+, $EMIM^+$, and $TBMP^+$) stabilized the negative charge located on the polymer backbone and prevented the chemical equilibrium from the doped state toward the undoped state. This was evidenced by an improved stability of electrical conductivity of the cation-exchange film compared to the TDAE-doped samples, as displayed in Figure S7. The cation-exchange doped films display nearly unchanged optical spectra within 20 h in an inert atmosphere.

Similarly, cation-exchange doping of f-BTI2g-SVSCN also showed improved thermoelectric performance compared to TDAE vapor doping, for which the $EMIM^+$ exchanged film exhibited the best electrical and thermoelectric performance (σ = 25.9 S cm^{-1}, S = −122 μV K^{-1}, and PF = 38.6 μW m^{-1} K^{-2}), but the $TBMP^+$ counterpart displayed the least enhanced thermoelectric performance (σ = 7.0 S cm^{-1}, S = −175 μV K^{-1}, and PF = 21.5 μW m^{-1} K^{-2}). The undeveloped electrical and thermoelectric properties of $TBMP^+$ exchanged f-BTI2g-SVSCN were mainly dictated by the low exchange efficiency, as evidenced by the FT-IR spectra. Because of the low exchange efficiency, the cation-exchange films did not exhibit improved stability of electrical performance compared to TDAE doping (Figure S7). The more pronounced attenuation of the Li^+ and $EMIM^+$ films was attributed to their better electrical properties. This was also evidenced by the enhanced absorption of the neural band after 24h, as shown in Figure S7.

We then employed EPR to detect the carrier concentration of TDAE-doped P-3O while exchanging the countercation with Li^+, $EMIM^+$, and $TBMP^+$, as displayed in Figure 3d. They all displayed pronounced EPR signals, indicating considerable polaron-generating efficiency upon cation exchange. The calculated carrier density corresponds to 1.06 ± 0.08 × 10^{20}, 1.02 ± 0.12 × 10^{20}, and 1.83 ± 0.06 × 10^{20} cm^{-3} for Li^+, $EMIM^+$, and $TBMP^+$, respectively, indicating a deeper doping level for $TBMP^+$ exchanged P-3O film. The calculated carrier density of $TBMP^+$-P-3O film was also consistent with the results of quantitative XPS (2 × 10^{20} cm^{-3}). The carrier concentration of FastHall was also in accordance with that of EPR, for which the cation exchange with $TBMP^+$ exhibited the higher carrier concentration, as displayed in Figure 3e. Moreover, the cation exchange with $TBMP^+$ also displayed the largest Hall mobility (4.7 ± 0.6 × 10^{-2} cm^2 V^{-1} s^{-1}) compared to the Li^+ (2.4 ± 0.4 × 10^{-2} cm^2 V^{-1} s^{-1}) and $EMIM^+$ (3.7 ± 0.3 × 10^{-2} cm^2 V^{-1} s^{-1}) exchanged analogues, indicating a weaker carrier-counterion interaction between the doped P-3O and $TBMP^+$. Temperature-dependent conductivity was utilized to characterize the carrier transport energy barrier of the doped P-3O films within the three kinds of counterions, as shown in Figure 3f. A decreased transport activation energy was observed from Li^+ to $EMIM^+$ and $TBMP^+$, indicating that facilitated charge-transport properties with $TBMP^+$ served as a counterion in the doped P-3O films. Therefore, the improved carrier concentration, the charge mobility, and the low energy barrier dictated the enhanced electrical conductivity for the $TBMP^+$ exchanged P-3O films.

3.4. Structural Variation of Cation-Exchange Doped Films. AFM was employed to evaluate how the cations

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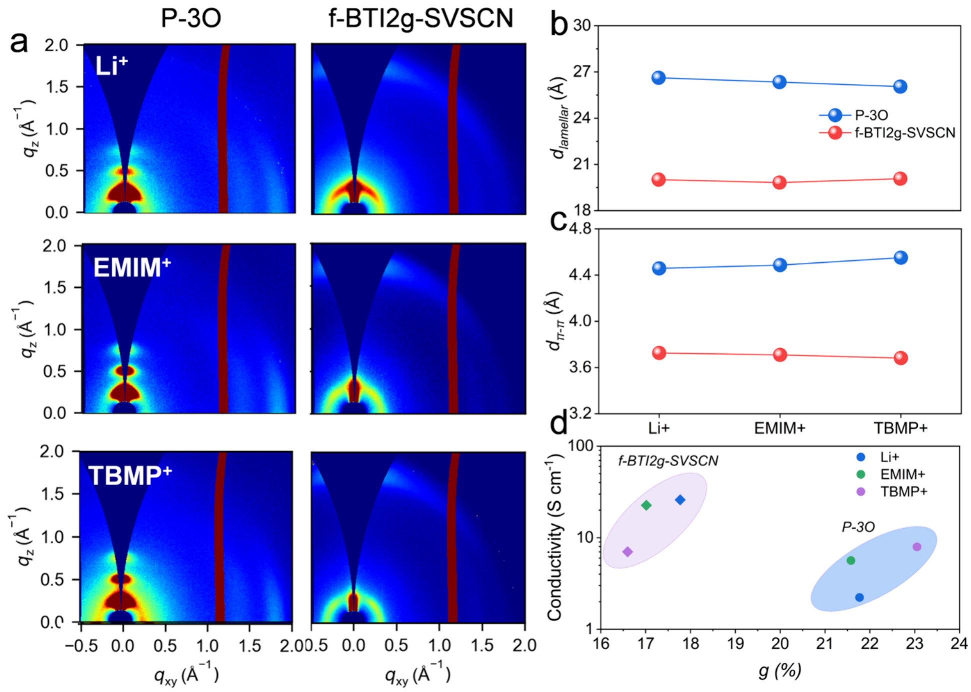

Figure 4. (a) GIWAXS 2D patterns, (b) π-stacking distance variation, and (c) lamellar packing space of Li^+, $EMIM^+$, and $TBMP^+$ exchanged P-3O and f-BTI2g-SVSCN. (d) Relationship between electrical conductivity and paracrystallinity.

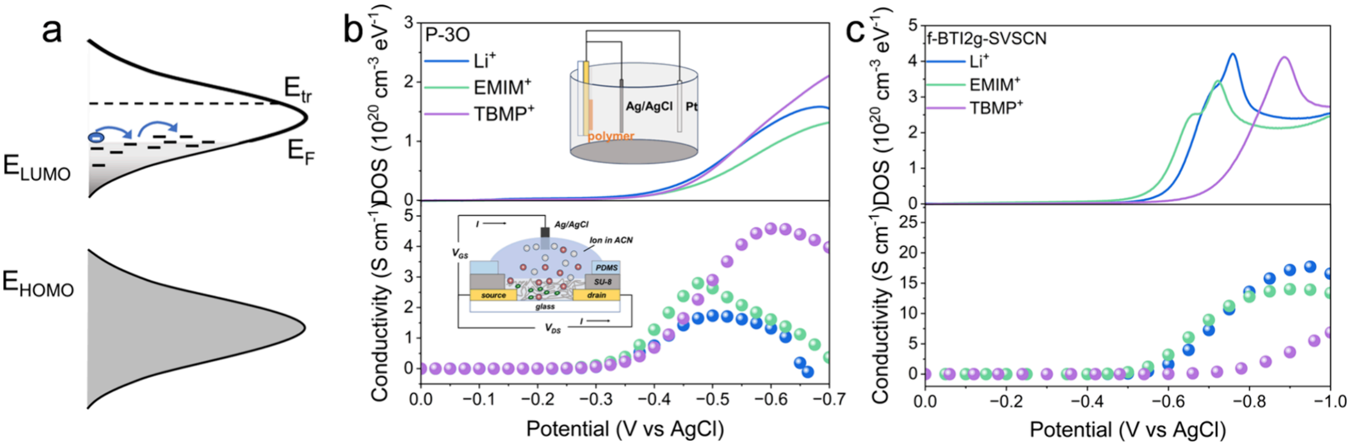

Figure 5. (a) Sketch map of DOS-determined charge transport. (b, c) Distribution (above) and gate voltage-mediated electrical conductivity (below) of P-3O and f-BTI2g-SVSCN films (LiTFSI, EMIMTFSI, and TBMPTFSI served as electrolytes, respectively.

influence the aggregates and surface morphology, as shown in Figure S8. The pristine P-3O film showed considerable molecule order with fabric aggregates. AFM exhibits nearly unchanged surface morphology with TDAE vapor doping and cation exchange, which might demonstrate an unchanged aggregating structure. The nearly uniform morphology was also observed in the cation exchange of f-BTI2g-SVSCN, as shown in Figure S8.

GIWAXS was adopted to investigate the crystalline structure variation during TDAE vapor doping and further cation exchange. The 2D patterns and linecuts in the direction of q_{xy} and q_z can be seen in Figures 4a and S9. P-3O adopted an edge-on predominant orientation, as evidenced by (100) (~0.24 A^{-1}) in the direction of q_z and (010) (~1.40 A^{-1}) in the q_{xy}. The high-order packings of (200) and (300) were also observed in q_z. Notably, the extra signal ~1.75 A^{-1} at the q_{xy} direction originated from the aggregate of thiazole. With cation exchange from Li^+ to $EMIM^+$ and $TBMP^+$, the lamellar packing distance shrank from 26.6 to 26.3 and 26.0 Å (Figure 4b). The π-stacking distance increased from 4.44 to 4.49 and 4.54 Å (Figure 4c). Considering the similar carrier concentrations evaluated by EPR and Fasthall measurements, this might indicate intercalation to lamellar packing and π-stacking for Li^+ and $TBMP^+$, respectively. f-BTI2g-SVSCN, which exhibited a face-on dominant orientation, displayed a totally different trend, where the π-stacking distance was slightly decreased from 3.72 to 3.70 and 3.68 Å for Li^+ to $EMIM^+$ and $TBMP^+$, respectively. The lamellar packing space was almost unchanged. However, by carefully analyzing the 2D GIWAXS patterns, the lamellar packing altered its preferred orientation from edge-on to mixed and face-on with cations varying from Li^+ to $EMIM^+$ and $TBMP^+$, respectively (the pristine one adopted a face-on dominant orientation). This might be evidence of a side chain-cation interaction and structural variation during cation exchange. Recently, excellent works addressed by Sirringhaus have denoted the correlation between paracrystallinity and the electrical conductivity of anion exchange in p-type doping.[38] However, there seems to be no strong correlation between electrical conductivity and paracrystallinity, no matter which packing orientation the polymer adopted, as the most conductive $TBMP^+$-P-3O (edge-on) and $EMIM^+$-f-BTI2g-SVSCN (face-on) displayed the most disordered π-stacking aggregates among other cation exchanged counterparts, as shown in Figure 4d. This might indicate that the π-stacking disorder-dominant electrical performance is unsuitable for describing the cation exchange in n-type polymers entailing glycolated side chains. Similarly, a recent work reported by Lei and co-workers demonstrated the noncovalent interaction between counterions and polymers but not paracrystalline disorder during cation-exchange doping.[53]

3.5. DOS Distribution and Electrochemically Modulated Conductivity. Herein, we proposed a new perspective to understand the cation-relative performance by the distribution of the DOS. The charge transport in organic semiconductors mainly adopts a hopping behavior on the unoccupied states near the Fermi level, which is highly correlated with the distribution of DOS, as illustrated in Figure 5a. The DOS is the number of states that electrons can occupy at a certain energy level. The distribution of the DOS indicates the paracrystalline disorder in the organic semiconductor film. Generally, a polymer film with a high disorder would exhibit a broad distribution of DOS, which is susceptible to the dopants and, thus, higher conductivity.[32,54,55] A narrow bandwidth of DOS endows the polymer film with a low charge-transport barrier and high carrier mobility.[18,56] The distribution of DOS could be monitored by directly detecting the abundance of electrochemically injected electrons in the materials.

The results are displayed in Figure 5b,c. The cations Li^+, $EMIM^+$, and $TBMP^+$ exhibited different effects on reducing P-3O and f-BTI2g-SVSCN. For P-3O, all three cations exhibit a similar onset reduction potential, while $TBMP^+$ displays a higher density of unoccupied state than the other two counterparts. According to the hopping transport model of organic semiconductors, the mobile carriers were hopping through unoccupied states near the Fermi level. This might explain why the TDAE-doped P-3O cation exchanged with $TBMP^+$ exhibits a higher carrier concentration than the other two. Further, gate potential mediated channel electrical conductivity was also studied, as shown in Figure 5b (below). A low sweeping speed of 1 mV s^{-1} was employed to be consistent with that of the DOS measurement. A small voltage difference of 10 mV between the drain and source electrodes was implemented to ensure the whole film was uniformly doped. The channel current loop was recorded with increasing gate potential (versus AgCl). The electrical conductivity with three cations exhibited a similar peak-like trend. This corresponds to a polaron−bipolaron transition with increasing carrier injection as the bipolaron presents decreased mobility compared to polaron. When it calls back to the results of Raman spectra, the cation-exchange doping strength corresponds to a gate voltage of −0.6 to −0.7 V. Compared to the analogues of Li^+ and $EMIM^+$, a higher electrical conductivity of $TBMP^+$ (~5 S cm^{-1}) was obtained at this voltage section, which is consistent with the DOS results. The discrepancy between the electrochemically mediated conductivity and molecule doping might mainly result from the swollen films of solvent ACN in electrochemical doping. In the case of f-BTI2g-SVSCN (Figure 5c), a good match between the DOS and the gate voltage-mediated electrical conductivity results was also observed. Notably, the film with $TBMP^+$ as electrolytes displayed a higher onset reduction voltage than Li^+ and $EMIM^+$. The overpotential needed for doping, that is, Gibbs free energy, in other words, might result from the structural variation induced by the $TBMP^+$ penetration. The large discrepancy of polarity between the branched glycol chain of f-BTI2g-SVSCN and the ethylene side chain of $TBMP^+$ shields the polymer from electrochemical doping and cation exchange with $TBMP^+$. Consequently, the gate potential affects electrical conductivity, and $TBMP^+$ exchanged f-BTI2g-SVSCN shows lower doping levels compared to Li^+ and $EMIM^+$. This highlights the importance of carefully selecting cations associated with the polymer for the n-type ion exchange.

4. CONCLUSIONS

This work presents a versatile cation-exchange doping strategy that can enhance the performance of n-type semiconductors in various polymer systems. We investigated two polymers with glycol side chains that have different packing orientations: P-3O and f-BTI2g-SVSCN. These polymers were treated with TDAE vapor doping and exchanged with three different cations, varying in ion size and polarity. To verify the cation exchange and doping level in the modified polymers, we employed several techniques, including UV−vis-IR spectroscopy, Raman spectroscopy, and XPS. Both the conjugated polymers demonstrated improved thermoelectric performance while maintaining a deep doping level over a short time frame. The efficiency of cation exchange varied depending on the polymer, and the electrical performance differed based on the type of cation used. Structural characterization using GIWAXS revealed that the cation exchange induced disorder in the polymer structure. Interestingly, the enhanced electrical conductivity did not have a straightforward correlation with the ordered π-stacking of the polymers. Further investigation into the DOS distribution and gate voltage-mediated electrical conductivity highlighted the relationship between the cation-related occupied DOS and the doping strength. This analysis helps explain the varying thermoelectric performance observed with different cation-exchange doping techniques.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.macromol.5c01513.

2D GIWAXS patterns of pristine P-3O and f-BTI2g-SVSCN films; normalized Raman spectra of electrochemically doped P-3O; Raman vibronic spectra; FT-IR spectra of TDAE and $TDAE^{\cdot+}$; optical spectra; electrical conductivity, Seebeck coefficient, and calculated power factor of TDAE vapor-doped P-3O; AFM images; and linecuts of 2D GIWAXS patterns (PDF)

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The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

TDAE,tetrakis(dimethylamino)ethylene; EMIM+,1-Ethyl-3-methylimidazolium; TBMP+,tributylmethylphosphonium; OSCs,organic semiconductors; ICT,integer charge transfer; PBTTT,poly[2,5-bis(3- tetradecylthiophen-2-yl)thieno[3,2-*b*]-thiophene]; UV–vis-IR,ultraviolet–visible-infrared spectroscopy; FT-IR,Fourier transform infrared spectroscopy; CF,chloroform; HFIP,hexafluoroisopropanol; ACN,acetonitrile; TFSI-,bis(trifluoromethylsulfonyl)imide; DOS,density of states; OTFTs,organic thin-film transistors; OECTs,organic electrochemical transistors

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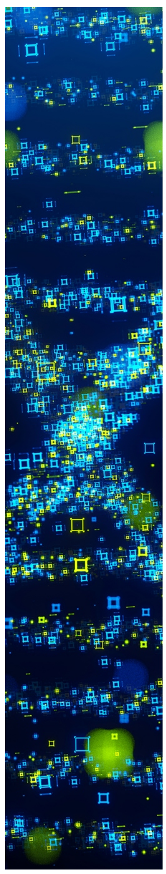

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